

Module 15: AKTU PYQs and Technical Interview Cheatsheet

1. Master Formula Sheet

Concept	Mathematical Formula
TLB EMAT	$h \times (t_{TLB} + t_{RAM}) + (1 - h) \times (t_{TLB} + 2 \times t_{RAM})$
Demand Paging EAT	$(1 - p) \times t_{RAM} + p \times t_{Fault}$
Segmentation	Physical = Base + Offset (If Offset < Limit)
Working Set Thrashing	Demand $D = \sum WSS_i > \text{Physical Memory } M$

Unit 4 High-Yield Formulas & Interview Cheatsheet

1. Paging & Address Translation

- Page Size = Frame Size = 2^n bytes $\Rightarrow$ Offset $d = n$ bits
- Page Number $p = \text{Logical Bits } (m) - n \text{ bits}$
- **Physical Address = (Frame Number $f \times \text{FrameSize}$) + d**
- Pure Paging Latency = $2 \times t_{RAM}$ (Double access penalty)
- Internal Frag = Worst (PageSize - 1), Average (PageSize / 2)

2. TLB & Effective Access Time

- Hit Time = $t_{TLB} + t_{RAM}$
- Miss Time = $t_{TLB} + 2 \times t_{RAM}$
- **EMAT = $h \times (t_{TLB} + t_{RAM}) + (1-h) \times (t_{TLB} + 2 \times t_{RAM})$**
- Two-Level Miss = $t_{TLB} + 3 \times t_{RAM}$
- Demand Paging EAT = $(1-p) \times t_{RAM} + p \times (t_{Fault})$

3. Page Replacement Rules

- FIFO: Replaces oldest; suffers Belady's Anomaly
- **OPT: Longest future use; minimum faults (theoretical)**
- **LRU: Least recent past use; Stack algorithm (NO anomaly)**
- Clock (Second Chance): $R=0$ replace; $R=1$ reset to 0
- Enhanced Clock: (0,0) best > (0,1) > (1,0) > (1,1) worst
- LFU/MFU: Smallest count / Largest count replaced

4. Segmentation & Thrashing

- Segment Address = $\langle s, d \rangle$ | If $d \geq \text{Limit} \Rightarrow$ Trap (Segfault)
- **If $d < \text{Limit} \Rightarrow \text{Physical Address} = \text{Base}(s) + d$**
- Thrashing: CPU thrashing when Demand $D = \sum WSS_i > M$
- PFF Strategy: Dynamic threshold frame allocation
- **Compaction: Requires Execution Time Dynamic Relocation**
- Cache AMAT = Hit_Time + Miss_Rate $\times$ Miss_Penalty

2. High-Yield Interview Topics

- **ASID Tagging vs TLB Flush:** Context switch par ASID tag use karke TLB flush avoid kiya jata hai.
- **Copy-On-Write (COW):** fork() par parent aur child shared read-only frames use karte hain; memory write par single page copy hota hai.
- **Belady's Anomaly:** FIFO suffers because it is not a stack algorithm ($S_n \not\subseteq S_{n+1}$).